

Basic Network Sniffer Report

Project Title

Basic Network Sniffer

Internship

CodeAlpha Cyber Security Internship

Developed By

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Objective

The objective of this project is to develop a Python-based Network Sniffer that captures live network traffic and analyzes packets to understand how data flows across a network. The project demonstrates the fundamentals of packet analysis and network protocols using the Scapy library.

Tools and Technologies

- Python 3
- Scapy Library
- Visual Studio Code
- Git & GitHub
- Windows Operating System

Project Description

The Network Sniffer captures live packets transmitted over the network and extracts useful information from each packet, including:

- Source IP Address
- Destination IP Address
- Network Protocol (TCP, UDP, ICMP)
- Packet Summary

The captured packet information is displayed in real time through the command-line interface, helping users understand network communication and packet structures.

Features

- Captures live network traffic
- Displays Source and Destination IP addresses
- Detects TCP, UDP, and ICMP protocols
- Displays packet summaries
- Provides real-time packet analysis

Working

1. Install Python and Scapy.
2. Run: **python network_sniffer.py**
3. The sniffer starts listening for live network packets.
4. Open websites such as Google or YouTube to generate traffic.
5. The program captures packets and displays their details in the terminal.
6. Stop the program by pressing Ctrl + C.

Sample Output

The program displays:

- Source IP Address
- Destination IP Address
- Protocol Type
- Packet Summary

Learning Outcomes

- Understanding packet sniffing
- Learning TCP/IP protocols
- Network traffic analysis
- Basics of cybersecurity monitoring
- Practical use of the Scapy library

Conclusion

The Basic Network Sniffer successfully captures and analyzes live network packets using Python and Scapy. This project provides practical knowledge of network communication, packet structures, and cybersecurity fundamentals, making it a valuable learning experience during the CodeAlpha Cyber Security Internship.